

CSC343H5 Introduction to Databases
Sprint 4: Normalization
 Student Submission Template

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How to use this template. Write each answer beneath its prompt, replacing the grey *[placeholder]* text. Use $\rightarrow$ for the arrow ($A \rightarrow B$), $\text{clo}\{\dots\}$ for a closure ($\{A\}^+$), and $\text{pk}\{\dots\}$ to underline a primary-key attribute. Show every step; closures and decomposition steps are where the marks are.

Reference: `research_flat`. Key (`grant_code`, `project_id`, `researcher_id`). Attributes: `grant_code`, `grant_title`, `funding_body`, `funding_sector`, `project_id`, `project_title`, `budget`, `researcher_id`, `researcher_name`, `institution`, `role_on_project`. Given FDs:

- FD1: `grant_code` $\rightarrow$ `grant_title`, `funding_body`
- FD2: `project_id` $\rightarrow$ `project_title`, `budget`
- FD3: `researcher_id` $\rightarrow$ `researcher_name`, `institution`
- FD4: `funding_body` $\rightarrow$ `funding_sector`
- FD5: `grant_code`, `project_id`, `researcher_id` $\rightarrow$ `role_on_project`

1. Task 1: Closure & Candidate Key

(a) Closure of the composite key

Let $K = \{\text{grant_code}, \text{project_id}, \text{researcher_id}\}$. Show each step; the final set must contain every attribute.

$$\begin{aligned}
 K^+ &= \{\text{grant_code}, \text{project_id}, \text{researcher_id}\} && \text{(start)} \\
 &= K^+ \cup \{\text{grant_title}, \text{funding_body}\} \\
 &\quad \text{(by FD1: } \text{grant_code} \rightarrow \text{grant_title}, \text{funding_body}\text{)} \\
 &= K^+ \cup \{\text{project_title}, \text{budget}\} \\
 &\quad \text{(by FD2: } \text{project_id} \rightarrow \text{project_title}, \text{budget}\text{)} \\
 &= K^+ \cup \{\text{researcher_name}, \text{institution}\} \\
 &\quad \text{(by FD3: } \text{researcher_id} \rightarrow \text{researcher_name}, \text{institution}\text{)} \\
 &= K^+ \cup \{\text{funding_sector}\} \\
 &\quad \text{(by FD4: } \text{funding_body} \rightarrow \text{funding_sector}, \text{ since funding_body is now in the set)} \\
 &= K^+ \cup \{\text{role_on_project}\} \\
 &\quad \text{(by FD5: } K \rightarrow \text{role_on_project}\text{)} \\
 &= \text{all 11 attributes} && \text{(done)}
 \end{aligned}$$

(b) Candidate key

$K = \{\text{grant_code}, \text{project_id}, \text{researcher_id}\}$ is the only candidate key. None of the other eight attributes (grant_title , funding_body , funding_sector , project_title , budget , researcher_name , institution , role_on_project) ever appears on the left-hand side of an FD, so no key can be built using them, and dropping any one of grant_code , project_id , researcher_id from K leaves the closure missing the attributes owned by the id that was dropped (e.g. $\{\text{project_id}, \text{researcher_id}\}^+$ never reaches grant_title or funding_body), so all three are required.

(c) Prime and non-prime attributes

Prime: grant_code , project_id , researcher_id **Non-prime:** grant_title , funding_body , funding_sector , project_title , budget , researcher_name , institution , role_on_project

2. Task 2: Minimal Cover**Step 1: Single-attribute right-hand sides**

Splitting each FD's right-hand side into single attributes gives 8 FDs:

- $\text{grant_code} \rightarrow \text{grant_title}$ $\text{grant_code} \rightarrow \text{funding_body}$
- $\text{project_id} \rightarrow \text{project_title}$ $\text{project_id} \rightarrow \text{budget}$
- $\text{researcher_id} \rightarrow \text{researcher_name}$ $\text{researcher_id} \rightarrow \text{institution}$
- $\text{funding_body} \rightarrow \text{funding_sector}$
- $\text{grant_code}, \text{project_id}, \text{researcher_id} \rightarrow \text{role_on_project}$

Step 2: Remove redundant left-hand-side attributes

Only FD5 ($\text{grant_code}, \text{project_id}, \text{researcher_id} \rightarrow \text{role_on_project}$) has more than one attribute on the left. We test whether each attribute can be dropped by computing the closure of the *other two* using the full FD set from Step 1:

$$\{\text{project_id}, \text{researcher_id}\}^+$$

reaches project_title , budget , researcher_name , and institution , but never role_on_project
 $\Rightarrow \text{grant_code}$ is **not** redundant.

$$\{\text{grant_code}, \text{researcher_id}\}^+$$

reaches grant_title , funding_body , funding_sector , researcher_name , and institution , but never role_on_project $\Rightarrow \text{project_id}$ is **not** redundant.

$$\{\text{grant_code}, \text{project_id}\}^+$$

reaches grant_title , funding_body , funding_sector , project_title , and budget , but never role_on_project $\Rightarrow \text{researcher_id}$ is **not** redundant. All three attributes are needed on the left of FD5, so nothing is removed in this step.

Step 3: Remove redundant FDs

For each FD $X \rightarrow Y$ from Step 1, we compute $\{X\}^+$ using the *other seven* FDs and check whether Y is still reached:

- Drop $\text{grant_code} \rightarrow \text{grant_title}$: $\{\text{grant_code}\}^+$ reaches only funding_body and $\text{funding_sector} \Rightarrow \text{grant_title}$ not reached, **keep**.
- Drop $\text{grant_code} \rightarrow \text{funding_body}$: $\{\text{grant_code}\}^+$ reaches only $\text{grant_title} \Rightarrow \text{funding_body}$ not reached, **keep**.
- Drop $\text{project_id} \rightarrow \text{project_title}$: $\{\text{project_id}\}^+$ reaches only $\text{budget} \Rightarrow \text{project_title}$ not reached, **keep**.
- Drop $\text{project_id} \rightarrow \text{budget}$: $\{\text{project_id}\}^+$ reaches only $\text{project_title} \Rightarrow \text{budget}$ not reached, **keep**.
- Drop $\text{researcher_id} \rightarrow \text{researcher_name}$: $\{\text{researcher_id}\}^+$ reaches only $\text{institution} \Rightarrow \text{researcher_name}$ not reached, **keep**.
- Drop $\text{researcher_id} \rightarrow \text{institution}$: $\{\text{researcher_id}\}^+$ reaches only $\text{researcher_name} \Rightarrow \text{institution}$ not reached, **keep**.
- Drop $\text{funding_body} \rightarrow \text{funding_sector}$: $\{\text{funding_body}\}^+$ reaches nothing new $\Rightarrow \text{funding_sector}$ not reached, **keep**.
- Drop FD5: without grant_code , project_id , $\text{researcher_id} \rightarrow \text{role_on_project}$, the closure of that same set still reaches every other attribute through FD1–FD4, but never $\text{role_on_project} \Rightarrow \text{keep}$.

No FD is redundant, so all 8 survive.

Resulting minimal cover

The minimal cover turns out to contain the same information as the original FD set in §3 (grouping same-left-hand-side FDs back together for readability):

$$\text{Minimal cover} = \left\{ \begin{array}{l} \text{grant_code} \rightarrow \text{grant_title, funding_body} \\ \text{project_id} \rightarrow \text{project_title, budget} \\ \text{researcher_id} \rightarrow \text{researcher_name, institution} \\ \text{funding_body} \rightarrow \text{funding_sector} \\ \text{grant_code, project_id, researcher_id} \rightarrow \text{role_on_project} \end{array} \right\}$$

3. Task 3: BCNF Decomposition

For each step, name the violating FD, give the two resulting relations with their keys, and show the lossless check. Add as many steps as you need.

Decomposition steps

Step 1. Decompose research_flat on $\text{grant_code} \rightarrow \text{grant_title, funding_body}$.

- **Violation:** grant_code is not a superkey of research_flat . $\{\text{grant_code}\}^+$ reaches only grant_title , funding_body , and (via FD4) funding_sector — it never reaches project_id

or `researcher_id`, so it does not reach all 11 attributes.

- **Resulting relations:** `R1(grant_code, grant_title, funding_body, funding_sector)` with key grant_code (`funding_sector` is included because `grant_code`'s closure reaches it transitively through `funding_body`); `R2(grant_code, project_id, project_title, budget, researcher_id, researcher_name, institution, role_on_project)` with key grant_code, project_id, researcher_id.
- **Lossless check:** shared attributes = {`grant_code`}; `grant_code` is a key of `R1`, so the split is lossless.

Step 2. Decompose `R1` on `funding_body` $\rightarrow$ `funding_sector`.

- **Violation:** `funding_body` is not a superkey of `R1`. $\{funding_body\}^+$ reaches only `funding_sector` — it never reaches `grant_code`, the key of `R1`.
- **Resulting relations:** `funder(funding_body, funding_sector)` with key funding_body; `grant(grant_code, grant_title, funding_body)` with key grant_code.
- **Lossless check:** shared attributes = {`funding_body`}; `funding_body` is a key of `funder`, so the split is lossless.

Step 3. Decompose `R2` on `project_id` $\rightarrow$ `project_title, budget`.

- **Violation:** `project_id` is not a superkey of `R2`. $\{project_id\}^+$ reaches only `project_title` and `budget` — it never reaches `grant_code` or `researcher_id`, part of `R2`'s key.
- **Resulting relations:** `project(project_id, project_title, budget)` with key project_id; `R2b(grant_code, project_id, researcher_id, researcher_name, institution, role_on_project)` with key grant_code, project_id, researcher_id.
- **Lossless check:** shared attributes = {`project_id`}; `project_id` is a key of `project`, so the split is lossless.

Step 4. Decompose `R2b` on `researcher_id` $\rightarrow$ `researcher_name, institution`.

- **Violation:** `researcher_id` is not a superkey of `R2b`. $\{researcher_id\}^+$ reaches only `researcher_name` and `institution` — it never reaches `grant_code` or `project_id`, part of `R2b`'s key.
- **Resulting relations:** `researcher(researcher_id, researcher_name, institution)` with key researcher_id; `participation(grant_code, project_id, researcher_id, role_on_project)` with key grant_code, project_id, researcher_id.
- **Lossless check:** shared attributes = {`researcher_id`}; `researcher_id` is a key of `researcher`, so the split is lossless.

No further FDs are violated in any of the five relations below, so the decomposition is complete.

Final BCNF schema

- `funder(funding_body, funding_sector)`
- `grant(grant_code, grant_title, funding_body)`
- `project(project_id, project_title, budget)`

- `researcher(researcher_id, researcher_name, institution)`
- `participation(grant_code, project_id, researcher_id, role_on_project)`

(`grant.funding_body` is a foreign key into `funder`; `participation`'s three key columns are foreign keys into `grant`, `project`, and `researcher` respectively.)

BCNF verification

In each relation, the left-hand side of the only non-trivial FD that applies is exactly that relation's primary key, so every non-trivial FD has a superkey left-hand side:

- `funder`: `funding_body` $\rightarrow$ `funding_sector`, and `funding_body` is the key.
- `grant`: `grant_code` $\rightarrow$ `grant_title`, `funding_body`, and `grant_code` is the key.
- `project`: `project_id` $\rightarrow$ `project_title`, `budget`, and `project_id` is the key.
- `researcher`: `researcher_id` $\rightarrow$ `researcher_name`, `institution`, and `researcher_id` is the key.
- `participation`: `grant_code`, `project_id`, `researcher_id` $\rightarrow$ `role_on_project`, and this combination is the whole key.

All five relations are in BCNF.

4. Task 4: 3NF Check & Justification

(a) Dependency preservation

Each FD in the minimal cover can be checked entirely inside one of the five final relations, with no join required:

- `grant_code` $\rightarrow$ `grant_title`, `funding_body` — checkable in `grant`.
- `project_id` $\rightarrow$ `project_title`, `budget` — checkable in `project`.
- `researcher_id` $\rightarrow$ `researcher_name`, `institution` — checkable in `researcher`.
- `funding_body` $\rightarrow$ `funding_sector` — checkable in `funder`.
- `grant_code`, `project_id`, `researcher_id` $\rightarrow$ `role_on_project` — checkable in `participation`.

Since every FD lands cleanly in exactly one relation, this BCNF decomposition is **dependency-preserving**. A 3NF synthesis is not needed here; the BCNF schema above can be used directly, since it gives both no redundancy (BCNF) and dependency preservation, which is not always possible together but happens to hold for this FD set.

(b) Anomaly example

Update anomaly. In `research_flat`, `grant NSERC-4471` appears on two rows: (`NSERC-4471`, `EcoForecast`, `Amara Osei`, `PI`) and (`NSERC-4471`, `EcoForecast`, `Priya Sharma`, `Postdoc`). The value of `grant_title` is copied onto both rows. If the grant is renamed, both rows must be updated; if only one is changed, the table now claims `NSERC-4471` has two different titles at once, an inconsistency that cannot happen in the decomposed schema, where `grant_title` is stored exactly once, in the `grant` table, keyed by `grant_code`.

5. AI Usage Declaration

I used an AI tool (Claude) while completing this sprint, as declared below.

- **Tool:** Claude Sonnet 5

What I asked: *[Describe in your own words, e.g. “I asked it to walk through the closure computation, minimal cover reduction, and BCNF decomposition steps for the given FD set, and explain the reasoning at each step.”]*

How I verified / applied it: *[Describe how YOU checked the work, e.g. “I re-computed each closure by hand from the FDs in §3 and confirmed the redundancy tests in Task 2 and the BCNF violation checks in Task 3 independently before including them here.”]*

6. Self-Check (optional)

- ☐ Closure reaches all 11 attributes.
- ☐ Every minimal-cover step shows a closure test.
- ☐ Each BCNF step has a violation, two relations with keys, and a lossless check.
- ☐ Every final relation is verified BCNF.
- ☐ Name, student number, and UTORid filled in above.